

TASK 9 – DESCRIPTIVE STATISTICS PRIMER

Titanic Dataset Analysis

1. Introduction

Descriptive statistics provide a concise summary of the main characteristics of a dataset. In this task, descriptive statistics were calculated for selected numeric variables from the Titanic dataset to understand central tendency, spread, percentiles, and skewness.

2. Objective

The objective of this task is to calculate and interpret mean, median, mode, standard deviation, percentiles, and skewness, and to build an understanding of what these statistics reveal about a distribution.

3. Dataset

The Titanic dataset was used for the analysis. Five numeric columns were selected: Age, Fare, SibSp (siblings/spouses aboard), Parch (parents/children aboard), and Pclass (passenger class). Missing observations were excluded column-wise when calculating the statistics.

4. Tools Used

Python, Pandas, NumPy, and Microsoft Excel were used to perform and present the analysis.

5. Descriptive Statistics Summary

Variable	Mean	Median	Mode	Std Dev	25th %	50th %	75th %	Skewness
Age	35.67	36.00	36.00	15.64	24.00	36.00	47.50	0.01
Fare	78.68	57.00	26.55	76.35	29.70	57.00	90.00	2.72
SibSp	0.46	0.00	0.00	0.64	0.00	0.00	1.00	1.44
Parch	0.47	0.00	0.00	0.75	0.00	0.00	1.00	1.52
Pclass	1.19	1.00	1.00	0.52	1.00	1.00	1.00	2.68

6. Interpretation of Results

Age: The mean age is 35.67 and the median is 36.00. Since the skewness is approximately 0.01, Age is approximately symmetric. The middle 50% of observations lies between 24.00 and 47.50 years.

Fare: Fare has a mean of 78.68 and a median of 57.00. Its skewness of 2.72 indicates strong positive/right skewness. A smaller number of high fares pull the mean upward. The standard deviation of 76.35 indicates substantial variation in fares.

SibSp: SibSp has a mean of 0.46 and a median of 0.00. Its skewness of 1.44 indicates positive/right skewness. Most passengers have zero or a small number of siblings/spouses aboard, while fewer passengers have larger values.

Parch: Parch has a mean of 0.47 and a median of 0.00. Its skewness of 1.52 indicates positive/right skewness. Most observations are concentrated at lower values, with fewer passengers having larger family counts.

Pclass: Pclass has a mean of 1.19 and a median of 1.00. Its skewness statistic is positive at 2.68. However, Pclass is a discrete ordinal variable, so skewness should be interpreted cautiously rather than as if it were a continuous measurement.

7. Mean vs. Median

Comparing mean and median helps identify skewness. When the mean is noticeably higher than the median, relatively high values may be pulling the average upward. For this dataset, Fare shows a clear difference between mean and median, consistent with its strong positive skewness. Median is generally less affected by extreme values and can therefore be useful for strongly skewed variables.

8. Standard Deviation and Spread

Standard deviation measures how widely observations are distributed around the mean. Fare has a comparatively large standard deviation of 76.35, indicating considerable variation in ticket fares. Standard deviation should be interpreted together with the mean and the units of the variable.

9. Percentiles and Middle 50%

The 25th and 75th percentiles define the range containing the middle 50% of observations. For example, Age has a 25th percentile of 24.00 and a 75th percentile of 47.50, so the middle half of the analyzed Age observations falls within that interval.

10. Conclusion

The descriptive statistics analysis provides a clear overview of the center, spread, and distribution of the selected Titanic variables. Age is approximately symmetric, while Fare, SibSp, and Parch are positively skewed. Percentiles describe the central range of observations, and standard deviation provides additional information about variation. The analysis demonstrates how descriptive statistics can be used as an initial step in understanding a dataset before further analysis or modeling.